

 EXECUTIVE BRIEFING

Autonomous & Responsible AI Collections Strategy

Scaling intelligent financial recovery through predictive risk scoring, agentic automation, and rigorous ethical guardrails.

PREPARED BY

Data & Analytics Team

FOCUS DOMAIN

Geldium Risk Operations

ARCHITECTURE FRAMEWORK

Agentic AI & Responsible Governance

End-to-End System Architecture

OPERATIONAL FLOW

1

Data Inputs

- **Credit Utilization:** Flags accounts exceeding 80% limit.
- **Payment Behavior:** Tracks missed payment milestones ($\geq \$5$).
- **Customer Profile:** Subprime score (400-450) & card type.
- **Interaction History:** Channel preferences & responses.



2

Predictive Logic

- **ML Risk Engine:** Calculates real-time default probability.
- **Segmentation:** Assigns risk tiers (Low, Medium, High).
- **SHAP Explainability:** Identifies top individual risk drivers.
- **Trigger System:** Generates next-best-action alerts.



3

Targeted Actions

- **Self-Service:** Automated SMS/Email balance warnings.
- **Payment Plans:** Dynamic micro-restructuring offers.
- **Human Escalation:** Routes complex cases to agents.
- **Hardship Assistance:** Empathetic support flows.



4

Learning Loop

- **Outcome Tracking:** Logs payment conversion rates.
- **Feedback Assimilation:** Records agent overrides.
- **Model Retraining:** Continuous monthly recalibration.
- **Audit Logging:** 100% compliance record keeping.

Role of Agentic AI: Autonomy vs. Oversight

OPERATING MODEL

Autonomous Execution (AI Agent)

- ✓ **Soft Balance Reminders:** Automated proactive notifications for early-stage accounts crossing 80% utilization.
- ✓ **Self-Service Restructuring:** Generating standardized 3 to 6-month payment plan options via mobile app/SMS portal.
- ✓ **Channel Optimization:** Dynamically choosing optimal outreach timing and channels based on response history.
- ✓ **Standard Inquiries:** Processing inbound promise-to-pay confirmations and routine balance checks instantly 24/7.

Human-in-the-Loop (HITL) Oversight

- ✓ **High-Value Hardship Cases:** Negotiating custom debt settlement terms for accounts exceeding \$25,000 balance.
- ✓ **Vulnerable Customer Reviews:** Manual review of complex financial hardship or dispute claims.
- ✓ **Legal & Write-off Approvals:** Approving formal default triggers, legal proceedings, or credit bureau escalations.
- ✓ **Model Override Audit:** Reviewing edge-case decisions flagged by governance triggers or agent overrides.

Responsible AI Guardrails

GOVERNANCE FRAMEWORK



1. Fairness & Bias Mitigation

Ensuring non-discriminatory predictive outcomes across protected demographic groups.

- ▶ Demographic Parity & Equalized Odds quarterly bias audits.
- ▶ Proxy variable scrubbing (Zip Code, employment proxies).
- ▶ Fairness re-weighting during model training cycles.



2. Explainability & Transparency

Providing clear, plain-language reasoning for all credit risk predictions and actions.

- ▶ SHAP values embedded into collector dashboards.
- ▶ Adverse action notices with concrete top-3 risk reasons.
- ▶ No "black-box" decisioning for loan modification denials.



3. Compliance & Security

Strict adherence to consumer finance regulations and data protection laws.

- ▶ Automated FDCPA & TCPA contact time/frequency limits.
- ▶ Real-time disclosure insertion on all AI communications.
- ▶ Full immutable audit logs for regulatory inspections.

Expected Business Impact & Value Creation

STRATEGIC OUTCOMES

Quantitative KPIs

-20%

90-Day Delinquencies in Target Portfolio

-35%

Cost-to-Collect per Dollar Recovered

- ✓ **+15% Increase in Promise-to-Pay Conversions:** Driven by timely, individualized payment plan suggestions.
- ✓ **50% Faster Resolution Cycle Time:** Instant digital self-service options accelerate account resolution.

Qualitative & Strategic Outcomes

- ✓ **Enhanced Customer Retention & Trust:** Proactive, non-punitive balance warnings preserve customer relationships before default occurs.
- ✓ **Frictionless Self-Service Hardship:** Empowers cardholders to resolve delinquencies discreetly without phone stress.
- ✓ **100% Audit Readiness:** Full transparency and compliance records safeguard Geldium against regulatory penalties.
- ✓ **Higher Collector Productivity:** Human specialists focus exclusively on high-touch, empathetic hardship negotiations.